genetic and somatic distance.[41] These two measures are again aggregated at the bank–target-country level by using bank branch networks as relative weights.

Table 5 shows that the results carry over. Columns (1) and (2) present 2SLS estimates for the full sample, and columns (3) and (4) for the sample where we drop banks' home country exposures. The F-statistics on the first stage are considerably above the Stock-Yogo threshold of 10. Both instruments are significant at conventional levels with expected signs (greater genetic or somatic distance influencing trust negatively). The coefficients on bank-level trust bias are consistently significant at the 1% level. They are larger than the baseline estimates in Table 2. The most conservative of these estimates (i.e., 2.037 in column (1) of panel B) indicates that a one -standard-deviation rise in bank-level trust bias increases the probability of investing in a target country by more than 18%. This corresponds to nearly half of the observed diversification gap in European banks' sovereign portfolios.

5.3. Trust and Cultural Distance

In light of the large literature on cultural distance and financial outcomes (e.g., Siegel et al., 2011; Ahern et al., 2015), one may ask whether our results are simply capturing the effects of cultural distance as operationalized by Hofstede (1980, 2001), Schwartz (1994), Gelfand et al. (2011), and Pellegrino, Spolaore, and Wacziarg (forthcoming) rather than identifying a distinct factor, namely trust.

In panel A of Table 6, we document the relationship between four frequently used measures of cultural distance and sovereign exposures. Column (1) utilizes the first principal component of the six cultural dimensions of Hofstede (1980, 2001).[42] Column (2) then utilizes the first principal component of the three dimensions identified by Schwartz (1994).[43] Column (3) uses the cultural tightness index of

41. Our bilateral country-level measure of genetic distance computes the probability that two random alleles taken from the DNA sequences of two different country populations do *not* overlap (Cavalli-Sforza, Menozzi, and Piazza, 1996). This measure is evolutionarily correlated with the length of time that passed since the two populations separated back in history. Country-level somatic distance is based on anthropometric measures on four dimensions in a population: height, cephalic index, hair color, and skin pigmentation (Biasutti, 1954). Countries are first divided into three ordinal categories in each dimension. Somatic distance for each country-pair is then computed by taking the sum of the absolute differences in each of these four dimensions. Both measures are provided by Guiso et al. (2009). Hence, the bilateral trust instrumented in these regressions derives from deeper ethnic and ancestral ties.

42. Of these six dimensions, Long-term Orientation defines cultures prioritizing practicality in the long term, whereas Individualism emphasizes cultures centered around self-sufficiency. Uncertainty Avoidance assesses an individual's unease with unpredictability and ambiguity, while Masculinity accentuates traits like competitiveness and assertiveness. Power Distance and Indulgence gauge the importance of hierarchy and of satisfaction of human desires in a society, respectively. We take the first principal component of these six dimensions to minimize multicollinearity in our regressions.

43. Schwartz (1994) distinguishes three dimensions of national culture: embeddedness, harmony, and egalitarianism. We take the first principal component of these three dimensions to minimize multicollinearity in our regressions.

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TABLE 5. *Instrumented* bank-level trust bias and probability of sovereign exposure.

Outcome →	(1) (2SLS) Sovereign exposure	(2) (First-stage) Bank-level trust bias	(3) (2SLS) Sovereign exposure	(4) (First-stage) Bank-level trust bias
Panel A				
Bank-level trust bias	3.014***		4.491***	
	[0.871]		[1.608]	
Bank-level branches	−0.167***	0.007	−1.113**	0.149**
	[0.055]	[0.013]	[0.453]	[0.066]
Bank-level branches (squared)	0.020**	−0.003	1.015*	−0.123
	[0.009]	[0.003]	[0.516]	[0.089]
Bank-level genetic distance		−13.676***		−7.619***
		[1.229]		[0.882]
First-stage Kleibergen–Paap F-stat		123.84		74.65
First-stage Montiel–Pflueger F-stat		178.86		74.18
Panel B				
Bank-level trust bias	2.037***		2.658***	
	[0.531]		[0.874]	
Bank-level branches	−0.164***	0.004	−0.786**	0.105**
	[0.055]	[0.010]	[0.396]	[0.045]
Bank-level branches (squared)	0.017*	−0.002	0.724	−0.081
	[0.010]	[0.003]	[0.459]	[0.061]
Bank-level somatic distance		−0.043***		−0.027***
		[0.003]		[0.003]
First-stage Kleibergen–Paap F-stat		221.00		69.15
First-stage Montiel–Pflueger F-stat		254.85		137.79
Bank × time FEs	Yes	Yes	Yes	Yes
Home-country × target-country × time FEs	Yes	Yes	Yes	Yes
Observations	21,615	21,615	20,241	20,241
Bank sample	All	All	Foreign	Foreign

Notes: The table summarizes the results of 2SLS estimations over the full sample period from 2010-Q1 to 2021-Q2. *Bank-level Trust Bias* is instrumented with *Bank-level Genetic Distance* in panel A and with *Bank-level Somatic Distance* in panel B. Dependent variable is *Sovereign Exposure*, defined as a dummy variable indicating any positive exposure of a bank toward a target country at a point in time reported in EBA and CEBS disclosures. *Bank-level Trust Bias* is computed for each bank–target-country pair as the branch-weighted average (see equation (2)) of the residuals from a gravity model of trust (see equation (3)), where trust is defined as the portion of individuals in home country expressing "a lot of trust" toward target country, measured via Eurobarometer surveys. *Bank-level Branches* measures the number of bank branches (in thousands) that the bank owns in the target country. All branch-related information is from SNL Financial. For the definition of the other variables, see Section 5.2. First-stage Montiel–Pflueger F-stat tests for weak instruments in the first-stage estimation are implemented in a way that is robust to heteroscedasticity, autocorrelation, and clustering. Robust standard errors are clustered at the bank level and reported in brackets. *Significant at 10%; **significant at 5%; ***significant at 1%.

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TABLE 6. Bank-level trust bias and probability of sovereign exposure when controlling for bank-level cultural distance.

Outcome →	(1) Sovereign exposure	(2) Sovereign exposure	(3) Sovereign exposure	(4) Sovereign exposure	(5) Sovereign exposure
Panel A					
Bank-level trust bias	1.562***				
	[0.310]				
Bank-level cultural distance by Hofstede		−0.116***			
		[0.024]			
Bank-level cultural distance by Schwartz			−0.066***		
			[0.023]		
Bank-level cultural distance by Gelfand et al.				−6.675	
				[4.898]	
Bank-level cultural distance by Pellegrino et al.					−8.269**
					[4.165]
Panel B					
Bank-level trust bias	1.562***	0.947**	1.101**	1.037**	1.287***
	[0.310]	[0.391]	[0.432]	[0.408]	[0.439]
Bank-level cultural distance by Hofstede		−0.069**	−0.073**	−0.070**	−0.075**
		[0.030]	[0.031]	[0.030]	[0.031]
Bank-level cultural distance by Schwartz			0.027	0.023	0.011
			[0.029]	[0.026]	[0.028]
Bank-level cultural distance by Gelfand et al.				−2.246	−2.036
				[4.283]	[4.207]
Bank-level cultural distance by Pellegrino et al.					6.716
					[5.861]
Control for bank-level branches	Yes	Yes	Yes	Yes	Yes
Control for bank-level branches (squared)	Yes	Yes	Yes	Yes	Yes
Bank × time FEs	Yes	Yes	Yes	Yes	Yes
Home-country × target-country × time FEs	Yes	Yes	Yes	Yes	Yes
Observations	21,615	21,615	21,615	21,615	21,615

Notes: The table summarizes the results of equation (1) estimated over the full sample period from 2010-Q1 to 2021-Q2. Dependent variable is *Sovereign Exposure*, defined as a dummy variable indicating any positive exposure of a bank toward a target country at a point in time reported in EBA and CEBS disclosures. *Bank-level Trust Bias* is computed for each bank–target-country pair as the branch-weighted average (see equation (2)) of the residuals from a gravity model of trust (see equation (3)), where trust is defined as the portion of individuals in home country expressing "a lot of trust" toward target country, measured via Eurobarometer surveys. *Bank Branches* measures the number of bank branches (in thousands) that the bank owns in the target country. All branch-related information is from SNL Financial. For the definition of the other variables, see Section 5.3. Robust standard errors are clustered at the bank level and reported in brackets. *Significant at 10%; **significant at 5%; ***significant at 1%.

Gelfand et al. (2011).[44] The final column employs the cultural distance measure derived

44. Based on earlier anthropological work on small societies by Pelto (1968), Gelfand et al. (2011) measure how strict a country' residents are with its cultural norms without explicitly defining what those norms are and construct an index indicating the cultural tightness (or inversely, looseness) of a nation. We directly use this index in our regressions.

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from World Value Surveys by Pellegrino et al. (forthcoming).[45] For each of these measures, we first calculate/obtain cultural distance for each country-pair, computing absolute differences (or directly taking the values in the case of Pellegrino et al. (forthcoming)) and then aggregating at bank–target-country level, using bank branch networks as relative weights.

Trust and cultural distance both matter: all four distance measures have their expected negative sign and are statistically significant at the 1% level, with the exception of the cultural distance measure of Gelfand et al. (2011). When we include these variables as controls (panel B), *Bank-level Trust Bias* still carries a large and significant coefficient. Evidently, standard cultural distance proxies between banks and target countries cannot *fully* explain the impact of trust on sovereign exposures. Trust continues to have an independent effect on banks' lending decisions to governments.

5.4. External Validity: A New Online Survey/Corporate Exposures

In the preceding, we use the most recent Eurobarometer survey of cultural trust perceptions, which dates from 1996. Previous literature takes culture as given and assumes that it is persistent over time (see Guiso et al., 2009). Still, one may worry that trust perceptions have changed significantly. We therefore conducted a new online survey across 30 European countries to shed light on the stability of bilateral trust over time and the external validity of our results.

The new survey was conducted from June through December, 2022, with the help of the survey company Respondi and its partners. Respondents were asked a set of basic demographic questions as well as their trust levels, using the exact same question-and-answer choices as in the Eurobarometer survey. Each country's survey was translated into the local language and was made nationally representative by imposing quotas on demographic characteristics taken from Gallup World Polls. Only respondents who satisfied these quotas and passed attention tests are included in the final sample.[46] Table 1 provides the relevant summary statistics for bank-level treatment. Online Appendix Table D.1 lists the set of countries covered.

We can compare the Eurobarometer surveys and our online survey by plotting them side-by-side. Figure 4, which shows trust levels (biases) in panel A (B), confirms

45. These authors define cultural distance for each country pair as the average expected disagreement on a question of the World Values Survey by two individuals randomly drawn from those two countries (disagreement = 1 for different answers, 0 for identical answers). Only questions that are available for each country in the pair are used to compute the distance value, which means the precision of the estimate differs across country-pairs. We employ the earliest year available (i.e., 1984) to obtain a time-invariant cross-section of cultural distance values for each country-pair in our sample.

46. Online Appendix Table D.2 shows the resulting estimates of trust levels between countries, where trust is defined as the portion of individuals in the home country expressing "a lot of trust" toward target country. In Online Appendix Table D.3, we compute the resulting biases in trust after taking into account home- and target-country fixed effects. While we report the estimates only for 15 countries in these tables for comparability with Eurobarometer, our new survey includes information for an additional 15 European countries, which we also utilize in our analysis.

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